



Study on improving the inner surface quality of laser powder bed fusion-fabricated Hastelloy X by electrochemical polishing

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ABSTRACT

Although laser powder bed fusion (LPBF) exhibits significant advantages in forming complex structures, its application in precision engineering is severely constrained by poor surface quality. This study aims to enhance the surface quality of LPBF-fabricated Hastelloy X through electrochemical polishing (ECP) using eco-friendly NaCl-aqueous and NaCl-ethylene glycol (NaCl-EG) electrolytes. The effects of electrolyte composition and current density on electrochemical dissolution behavior were experimentally investigated. ECP parameters were optimized through a comprehensive evaluation of surface roughness, material removal rate (MRR), and thickness reduction. The results indicate that the formation of a supersaturated salt film in the NaCl-EG electrolyte can suppress the uneven dissolution of carbides and matrix, but its improvement on roughness is limited. Compared with the NaCl-EG electrolyte, the NaCl-aqueous electrolyte achieves comparable surface roughness while increasing the MRR by at least 336.21 %. Furthermore, to address the challenge of polishing internal cavity surfaces of LPBF parts, a conformal tool electrode design strategy is proposed. Through this approach, partially melted particles on the inner surface of curved channels were successfully removed, reducing the surface roughness S_a from 7.16 to 11.91 μm to 1.63–1.95 μm . This extends the application potential of ECP for polishing complex internal channels in LPBF components.

1. Introduction

Laser powder bed fusion (LPBF) is a high-precision metal additive manufacturing technology characterized by layer-by-layer powder deposition and selective laser melting to achieve near-net-shape forming of complex structural components [1]. Compared with traditional subtractive manufacturing processes, LPBF exhibits significant advantages including exceptional design freedom, high material utilization rate, and capability for integrated fabrication of topology-optimized structures, demonstrating substantial application potential in aerospace, automotive industries, and related fields [2].

Hastelloy X, a solution-strengthened nickel-based superalloy, has emerged as a critical material for manufacturing high-performance equipment such as aero-engine combustion chambers and gas turbine hot-section components, owing to its exceptional oxidation resistance, outstanding thermal corrosion resistance, and favorable high-temperature mechanical strength [3]. Leveraging its geometric freedom advantages, LPBF provides an innovative solution for Hastelloy

X in the fabricating of functional components with complex internal channel structures such as servo system precision actuators and fuel injection systems. Recent studies demonstrate that process optimization significantly enhances the density and mechanical properties of LPBF Hastelloy X. Han et al. [4] achieved components with density up to 99.3 % using optimized LPBF parameters. Murr et al. [5] and Etter et al. [6] further eliminated pores, microcracks, and reduced microstructural/mechanical anisotropy through hot isostatic pressing (HIP) treatment. However, the inherent step effect, balling phenomenon, and powder adhesion in LPBF processes result in poor surface quality, severely constraining engineering applications of this technology [7]. Firstly, stress concentrations are prone to form at surface roughness peaks and valleys, initiating fatigue cracks under cyclic loading. Secondly, irregular surfaces alter hydrodynamic characteristics, causing abnormal pressure fluctuations in systems. Most critically, the detachment of partially melted particles readily contaminates media and clogs microchannels, compromising component service reliability. Therefore, improving the surface quality of LPBF components through

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post-processing holds critical significance for promoting the use of LPBF for the engineering production of high-performance parts.

However, conventional mechanical polishing, while operationally simple and versatile, struggles to effectively process complex geometries (particularly high-aspect-ratio internal cavities) due to its contact-based nature [8]. Fluid-based chemical polishing offers superior accessibility, yet its efficacy heavily depends on solution composition, with existing polishing systems exhibiting inadequate compatibility for Ni-CrMo alloys [9]. In contrast, electrochemical polishing (ECP) is widely employed to enhance the surface quality of LPBF parts—including 316 L stainless steel [10], Ti-6Al-4V alloy [11], CoCrFeMnNi high-entropy alloy [12], and Inconel 625 [13]—owing to its capability to achieve nanoscale surface roughness. Furthermore, ECP enables surface leveling through controlled anodic dissolution without inducing mechanical damage or altering material properties [14]. Consequently, ECP has emerged as one of the most promising approaches for improving the surface quality of LPBF parts.

Conventional ECP typically operates at low current densities ($< 0.1 \text{ A}\cdot\text{cm}^{-2}$), relying on the formation of a viscous diffusion layer on the workpiece surface to achieve selective dissolution of micro-protrusions, thereby obtaining smooth surfaces [15]. However, this seems to be difficult to meet the demand for high polishing efficiency of LPBF parts, because there are a large number of partially melted particles and raised folds on the surface of LPBF parts, requiring a higher material removal rates (MRR). To this end, Chang et al. [16] proposed an overpotential electrochemical polishing method, which applies a slight overpotential to enable higher current densities and consequently elevate MRR. And the effectiveness of this method was confirmed in experiments on 316 L stainless steel. Furthermore, the electrolyte used in traditional ECP is usually acidic solutions such as phosphoric acid and sulfuric acid, which posing significant environmental and health challenges [17]. This reality has motivated researchers to develop eco-friendly electrolyte systems. Zhang et al. [18] investigated the dissolved surface quality of LPBF Hastelloy X in NaNO_3 and NaCl solutions, revealing that the corrosive effect of Cl^- effectively disrupts passive films, enhancing electrochemical dissolution uniformity. And Han et al. [19] demonstrated that 316 L stainless steel polished in NaCl -ethylene glycol (NaCl -EG) electrolyte achieves lower surface roughness and higher MRR compared to sulfuric acid-based electrolyte. These studies confirm the technical viability of replacing acidic electrolytes with neutral salt solutions for ECP of LPBF components. Nevertheless, despite successful applications in some alloy systems, reports on LPBF Hastelloy X remain scarce, with their dissolution behavior and polishing mechanisms still unclear and warranting further investigation.

Moreover, applying ECP to internal surfaces presents significantly greater challenges than external surfaces due to stringent requirements for tool electrode accessibility and electric field uniformity. To date, various forms of flexible tool electrodes have been tried for ECP of internal features in LPBF components. For instance, An et al. [20] engineered a cathode for polishing curved channels, comprising two or more twisted metallic wires interspersed with bundles of non-metallic microfilaments. Similarly, Zhao et al. [21] used a discretely insulated flexible cathode traversing bent internal channels in reciprocating motion to remove adhered powder from contoured surfaces. Nevertheless, effective methodologies for highly irregular internal geometries remain underdeveloped, representing a critical research challenge we aim to address.

In this work, the polishing effect of the inner surface of LPBF Hastelloy X parts in eco-friendly NaCl -aqueous and NaCl -EG electrolytes was systematically studied for the first time, and the electrochemical dissolution behavior and leveling process of LPBF Hastelloy X during ECP process was discussed. Furthermore, to address the persistent challenge of poor tool accessibility for internal surface polishing, we propose an integrated conformal tool electrode design strategy—fabricated concurrently with the component via LPBF—and demonstrate its successful implementation for ECP of curved channels,

effectively reducing surface roughness.

2. Material and experimental procedures

2.1. Sample preparation

In this study, Hastelloy X powder prepared by gas atomization method was used as the raw material for the LPBF process. Before LPBF, the powder underwent pretreatment in a vacuum drying oven at 120°C for 2 h. The key process parameters of LPBF were set as follows: the laser power was 316 W, the scanning speed was 935 mm/s, the hatch spacing was 110 μm , and the layer thickness was 50 μm . During the LPBF process, high-purity argon gas was introduced into the molding chamber to maintain an inert protective atmosphere. As shown in Fig. 1, three Hastelloy X samples with different structures were designed and prepared. Among them, the cubic samples (Fig. 1a) were used for the test of polarization curve, electrochemical impedance spectroscopy (EIS) and current efficiency to evaluate the electrochemical behavior of LPBF Hastelloy X. The Straight channel samples (Fig. 1b) were used for ECP parameters optimization in different electrolytes. In addition, synchronously formed curved channel samples with conformal tool electrode to validate internal ECP feasibility for complex geometries, as shown in Fig. 1c. And the conformal tool electrode maintained a precise inter-electrode gap with the channel's internal surface, enabling uniform polishing. All samples were treated with HIP to eliminate internal pores and microcracks generated during fabrication.

2.2. Electrochemical tests

Electrochemical tests, including polarization curve and EIS, were conducted in a three-electrode system using an electrochemical workstation (Zennium E, Zahner Instruments Inc.). A planar platinum electrode and a saturated calomel electrode (SCE) were served as the counter electrode and the reference electrode, respectively. As the working electrode, the LPBF Hastelloy X cubic sample has only one surface (1 cm^2) exposed to the electrolyte, while the other surfaces are covered with insulated epoxy resin. Before testing, the working electrode was soaked in the electrolyte for 2 h to stabilize the open circuit potential. EIS measurements were performed over a frequency range of 10^{-2} to 10^5 Hz with an amplitude of 10 mV. Acquired data were fitted using ZSimWin software. Polarization curves were recorded at a scan rate of 2 mV/s within a potential window of -2 –2 V. And the current efficiency (η) was determined using the following equation [22]:

$$\eta = \Delta M / (KIt) \quad (1)$$

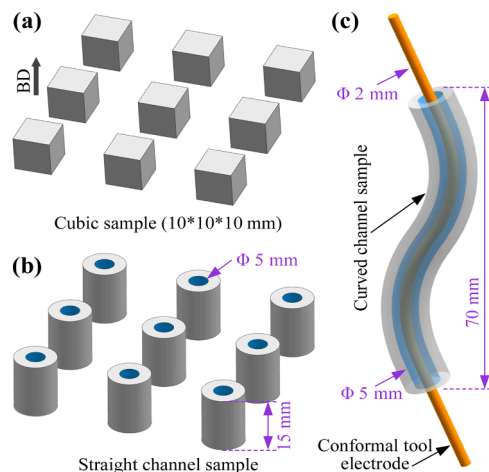


Fig. 1. Schematic illustration of (a) the cubic sample; (b) the straight channel sample and (c) the curved channel sample.

where ΔM is the dissolved mass (g); I is the current value (A); t is the dissolution time (s); and K is the theoretical value of the mass electrochemical equivalent ($\text{g} \cdot (\text{A} \cdot \text{s})^{-1}$). All tests and experiments were performed in 5 wt% NaCl-aqueous electrolyte and 5 wt% NaCl-EG electrolyte, which were obtained from NaCl (analytical grade) dissolved in deionized water and ethylene glycol (analytical grade) respectively.

2.3. Electrochemical polishing experiment

During the ECP process, the workpiece and tool electrode were accurately positioned and fixed through special fixtures. The electrolyte was continuously circulated through the channel via a pump to ensure dynamic renewal. For straight channels, a cylindrical 316 L stainless steel electrode with a diameter of 2 mm served as the tool electrode, and key parameters such as voltage and polishing time are optimized through process experiments.

In addition, the polishing procedure for curved channels is illustrated in Fig. 2. Initially, the workpiece served as the anode and the conformal tool electrode as the cathode to polish the inner surface of the workpiece. Upon completion of polishing, the conformal tool electrode was dissolved via electrode polarity reversal (cathode-anode switching), ultimately yielding an undamaged target component.

2.4. Surface characterization

After ECP, samples were sectioned along the central plane for observation and analysis. Surface morphology was characterized using scanning electron microscopy (SEM, Regulus8100, Hitachi) and the material composition was analyzed via energy dispersive spectroscopy (EDS). Critical roughness parameter S_a was quantitatively evaluated by laser confocal microscopy (OLS5100, Olympus), with all roughness data acquired within a fixed measurement area of $1280 \times 1280 \mu\text{m}$.

3. Results and discussion

3.1. Surface morphology and microstructure

In ECP, the polished surface quality of LPBF components is co-determined by the hereditary characteristics of the initial topography and the influence of microstructure on dissolution behavior. As illustrated in Fig. 3a, the initial surface of the sample exhibits typical LPBF process characteristics. Due to the steep thermal gradient at melt pool peripheries, surrounding powder only partially melt and adhere to the surface, forming abundant partially melted particles [23]. These particles exhibit high susceptibility to detachment owing to inferior bonding strength with the substrate, thereby constituting one of the most critical surface defects targeted for elimination via ECP. Furthermore, the

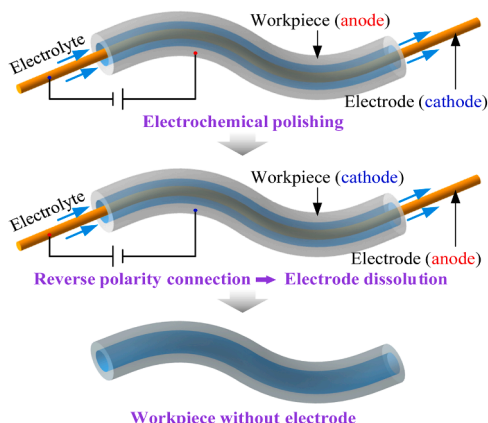


Fig. 2. Schematic diagram of ECP and conformal tool electrode removal.

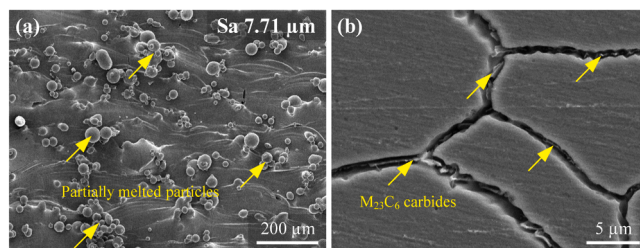


Fig. 3. The initial surface morphology (a) and microstructure characteristics (b) of LPBF Hastelloy X.

layer-wise fabrication mechanism generates wavelike protrusions across the substrate surface, which collectively with partially melted particles result in an initial surface roughness S_a as high as $7.71 \mu\text{m}$.

The microstructure of LPBF Hastelloy X is depicted in Fig. 3b, revealing continuously distributed precipitates along grain boundaries. Similar phenomena were observed by Lee et al. [24] and Kong et al. [25]. During LPBF, rapid cooling of the melt pool restricts the diffusion of solute atoms (e.g., C, Cr, Fe, Mo), resulting in the formation of a supersaturated solid solution. Following HIP, supersaturated solutes precipitate as M_{23}C_6 carbides at grain boundaries due to equilibrium solubility limitations. These carbides exhibit superior corrosion resistance compared to the matrix, potentially inducing inhomogeneous electrochemical dissolution behavior [22].

3.2. Electrochemical dissolution behavior

3.2.1. Electrochemical characteristics

To investigate the influence of electrolyte on the electrochemical characteristics of LPBF Hastelloy X, polarization curves were tested for the LPBF Hastelloy X in NaCl-aqueous and NaCl-EG electrolytes, respectively. As illustrated in Fig. 4, the LPBF Hastelloy X exhibited similar "passivation-transpassivation" behavior in both NaCl-aqueous and NaCl-EG electrolytes. This indicates that within the passivation potential range, a passive film formed on the LPBF Hastelloy X in both electrolytes, inhibiting further electrochemical dissolution. However, key parameters characterizing the performance of the passive film—transpassivation potential (E_{trans}) and passivation current density (i_{pass})—demonstrated significant differences between the two electrolytes. Generally, a lower E_{trans} implies that the passive film is more susceptible to breakdown. Research by Yin et al. [26] revealed that the primary constituents of the passive film on the surface of LPBF Hastelloy X are oxides and hydroxides of metallic elements. However, few existing studies have elucidated the formation mechanisms of passive films in NaCl-EG electrolyte. Guo et al. [27] proposed that ethylene glycol absorbs trace amounts of water from the atmosphere, thereby supplying

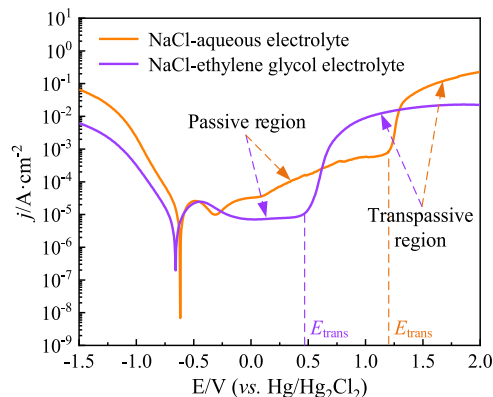


Fig. 4. Polarization curves of LPBF Hastelloy X in NaCl-aqueous and NaCl-EG electrolytes.

oxygen for passive film formation. Consequently, the lower oxygen content in ethylene glycol results in thinner passive films, rendering them more susceptible to breakdown. Furthermore, during electrochemical reactions, ethylene glycol decomposes into organic acids such as glycolic acid, acetic acid, oxalic acid, and formic acid [28]. The acidic environment further reduces the potential required for passive film breakdown. On the other hand, a lower i_{pass} corresponds to slower corrosion rates. Thus, passive LPBF Hastelloy X exhibits superior corrosion resistance in NaCl-EG electrolyte. One plausible explanation is that passive films formed in different electrolytes possess distinct properties, with the ethylene glycol-derived film displaying higher compactness, which reduces ion diffusion rates through film defects. In the transpassive region, passive film breakdown occurs, accompanied by rapid increases in current density and anodic dissolution rates. The higher current density in NaCl-aqueous electrolyte compared to NaCl-EG electrolyte can be ascribed to its superior solution conductivity.

To further investigate the properties of passive films of LPBF Hastelloy X in different electrolytes, EIS was tested in NaCl-aqueous electrolyte and NaCl-EG electrolyte, and the results were fitted and analyzed. In Nyquist diagram, a larger arc radius indicates greater charge transfer resistance. As shown in Fig. 5a, the arc radius of the LPBF Hastelloy X in NaCl-EG electrolyte is significantly larger than that in NaCl-aqueous electrolyte. Additionally, Fig. 5b and c display the bode phase and impedance diagram of the LPBF Hastelloy X in different electrolytes, respectively. At a frequency of 10^{-2} Hz, the phase angles of LPBF Hastelloy X in NaCl-aqueous and NaCl-EG electrolytes are -36° and -75° respectively, while the impedance $|Z|$ are $1.29 \times 10^6 \Omega \cdot \text{cm}^2$ and $3.56 \times 10^6 \Omega \cdot \text{cm}^2$, respectively. Generally, higher phase angles and impedance magnitudes in the low frequency region correspond to superior corrosion resistance of passive films.

To quantitatively analyze the performance differences of passive films in different electrolytes, the EIS data were fitted using equivalent

circuits. Considering the relaxation time distribution caused by the surface inhomogeneity of the passive film, a constant phase element (CPE) was used to describe the non-ideal capacitance behavior, and its definition given by Eq. (2) [22]:

$$Z_{\text{CPE}} = [Q(i\omega)^n]^{-1} \quad (2)$$

where Q is the CPE constant, ω is the angular frequency, i is the imaginary number ($i^2 = -1$), and n is the CPE exponent. The equivalent circuit is illustrated in Fig. 5d, where R_s , R_f , and R_{ct} represent the electrolyte resistance, passive film resistance, and charge transfer resistance, respectively, and Q_1 and Q_2 denote the passive film capacitance and interfacial double-layer capacitance, respectively. The fitting results are listed in Table 1. The Chi-square values (χ^2) are all below 10^{-3} , and the fitted curves exhibit close agreement with the experimental data, demonstrating the validity of the equivalent circuit. R_f and R_{ct} are inversely proportional to the corrosion rate and thus serve as indicators of corrosion resistance. The R_f and R_{ct} values of LPBF Hastelloy X in NaCl-EG electrolyte are significantly higher than those in NaCl-aqueous

Table 1

Simulation values of EIS for LPBF Hastelloy X in NaCl-aqueous and NaCl-EG electrolytes.

	NaCl-EG	NaCl-aqueous
$R_s (\Omega \cdot \text{cm}^2)$	1.66×10^2	15.28
$Q_1 (\text{F} \cdot \text{cm}^{-2})$	5.59×10^{-6}	2.21×10^{-5}
n_1	0.90	0.89
$R_f (\Omega \cdot \text{cm}^2)$	1.64×10^6	1.69×10^5
$Q_2 (\text{F} \cdot \text{cm}^{-2})$	1.25×10^{-4}	3.53×10^{-6}
n_2	0.91	0.92
$R_{ct} (\Omega \cdot \text{cm}^2)$	3.03×10^{10}	3.86×10^7
χ^2	8.71×10^{-5}	2.95×10^{-4}

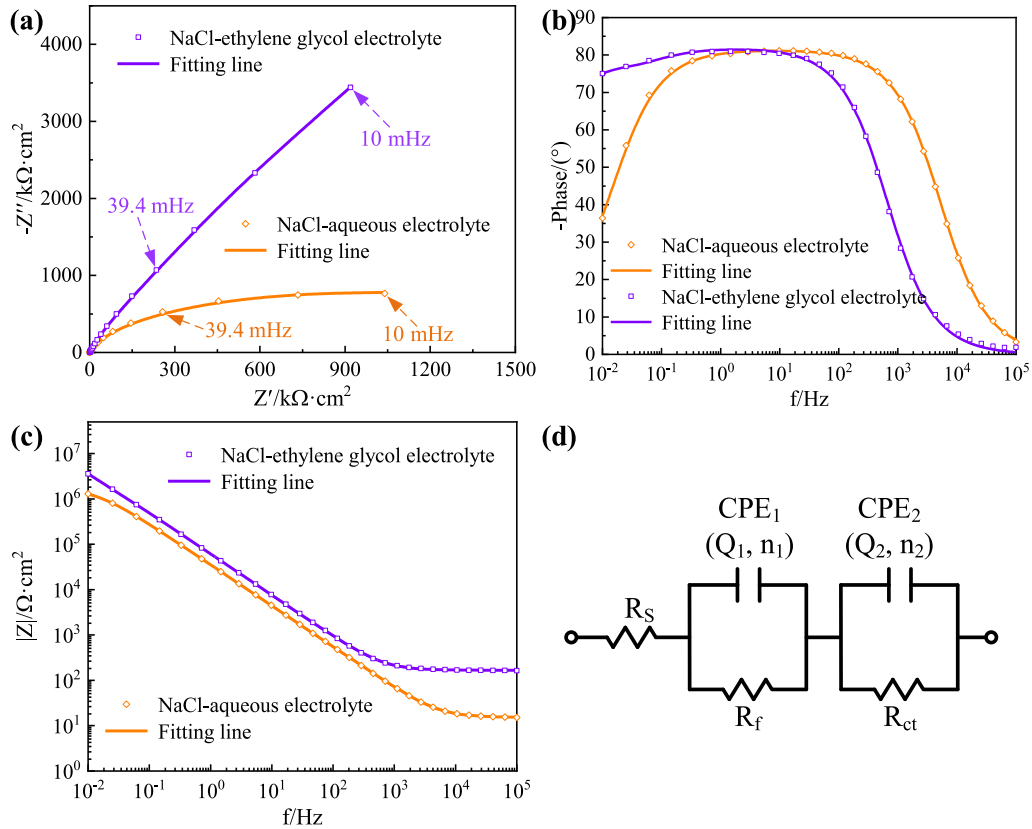


Fig. 5. EIS results of LPBF Hastelloy X in NaCl-aqueous and NaCl-EG electrolytes: (a) Nyquist diagram; (b) Bode phase diagram; (c) Bode impedance diagram; (d) equivalent circuit model.

electrolyte, which aligns with the polarization curve measurements.

3.2.2. Effect of electrolytes on dissolution behavior

To investigate the dissolution process of LPBF Hastelloy X in different electrolytes, electrochemical dissolution experiments were conducted at a current density of $1 \text{ A} \cdot \text{cm}^{-2}$ for varying durations, followed by surface morphology observations. Fig. 6a displays the surface morphology of LPBF Hastelloy X after 3 s of electrochemical dissolution in NaCl-aqueous electrolyte. It is clearly observed that dissolution preferentially initiates at grain centers, indicating localized breakdown of the passive film in these regions, thereby triggering non-uniform dissolution. When the dissolution time extends to 20 s, white protrusions appear near grain boundaries, as shown in Fig. 6b. EDS analysis was performed at two locations: Point 1 and Point 2, with results presented in Fig. 6d and e, respectively. Compared to Point 1, Point 2 exhibits significantly elevated C and O concentrations, confirming that the protrusions consist of carbides and oxides. Carbides located at grain boundaries exhibit higher dissolution resistance than the matrix material, thus, they become progressively exposed and form protrusions as the matrix dissolves. Additionally, Oxide formed by adsorption of electrolytic products and passive film further enhance the corrosion resistance of these regions. As dissolution continues to 40 s (Fig. 6c), the height difference between carbide protrusions and the matrix reaches $\sim 2.5 \mu\text{m}$, as shown in Fig. 6c.

Based on the above observations and analyses, an electrochemical dissolution model was established to elucidate the dissolution process of LPBF Hastelloy X in NaCl-aqueous electrolyte, as illustrated in Fig. 7. Fig. 7a depicts the initial dissolution stage of LPBF Hastelloy X. A passive film forms on the alloy surface, but the difference in composition between matrix and carbides at grain boundaries leads to the difference in the performance of passive film, and the passive film at grain boundaries is more stable. Under applied potential, the passive film on the matrix surface is preferentially breached, as shown in Fig. 7b. Subsequently, uneven electrochemical dissolution between the matrix and carbides

generate an uneven dissolved surface (Fig. 7c). Owing to their slower dissolution rate compared to the matrix, carbides become progressively exposed and form protrusions as the matrix dissolves. Additionally, electrolytic products adsorb onto the protruding carbides, these products serve as precursors that ultimately form the passive film, further inhibiting carbide dissolution [29]. However, the height differential between the matrix and carbides arising from inconsistent dissolution rates does not increase indefinitely. According to the distribution law of interelectrode electric fields, current density intensifies at protrusions. As the height of carbide protrusions increases, the dissolution rates of the matrix and carbides converge when electrochemical kinetics equilibrate with the applied electric field. At this point, the system transitions to a steady-state dissolution regime, as presented in Fig. 7d.

Fig. 8 exhibits the dissolved surface of LPBF Hastelloy X in NaCl-EG electrolyte, revealing fundamentally distinct behavior from that in NaCl-aqueous electrolyte. In Fig. 8a, no residual passive film is observed at grain boundary regions. Electrochemical test results indicate that the passive film formed on LPBF Hastelloy X in NaCl-EG electrolyte is relatively thin, which may account for its uniform breakdown. When dissolution duration extends to 20 s (Fig. 8b) and 40 s (Fig. 8c), the dissolved surfaces remain flat and smooth, signifying uniform dissolution of both the matrix and carbides in LPBF Hastelloy X within NaCl-EG electrolyte.

The electrochemical dissolution model of LPBF Hastelloy X in NaCl-EG electrolyte is illustrated in Fig. 9. In NaCl-EG electrolyte, the anodic and cathodic reactions can be expressed as Eqs. (3) and (4), respectively. Subsequently, cations (M^{n+}) combine with anions ($\text{HOCH}_2\text{CH}_2\text{O}^-$) to form soluble metal alkoxides, as shown in Eq. (5) [30]. Due to the high viscosity of ethylene glycol, the metal alkoxides near the anode surface cannot diffuse away promptly, forming a supersaturated salt film (Fig. 9a). This supersaturated salt film modifies the local current density distribution at the anode surface, thereby promoting surface smoothing. Moreover, it dominates the electrochemical dissolution behavior, suppressing selective dissolution caused by inhomogeneities in the passive

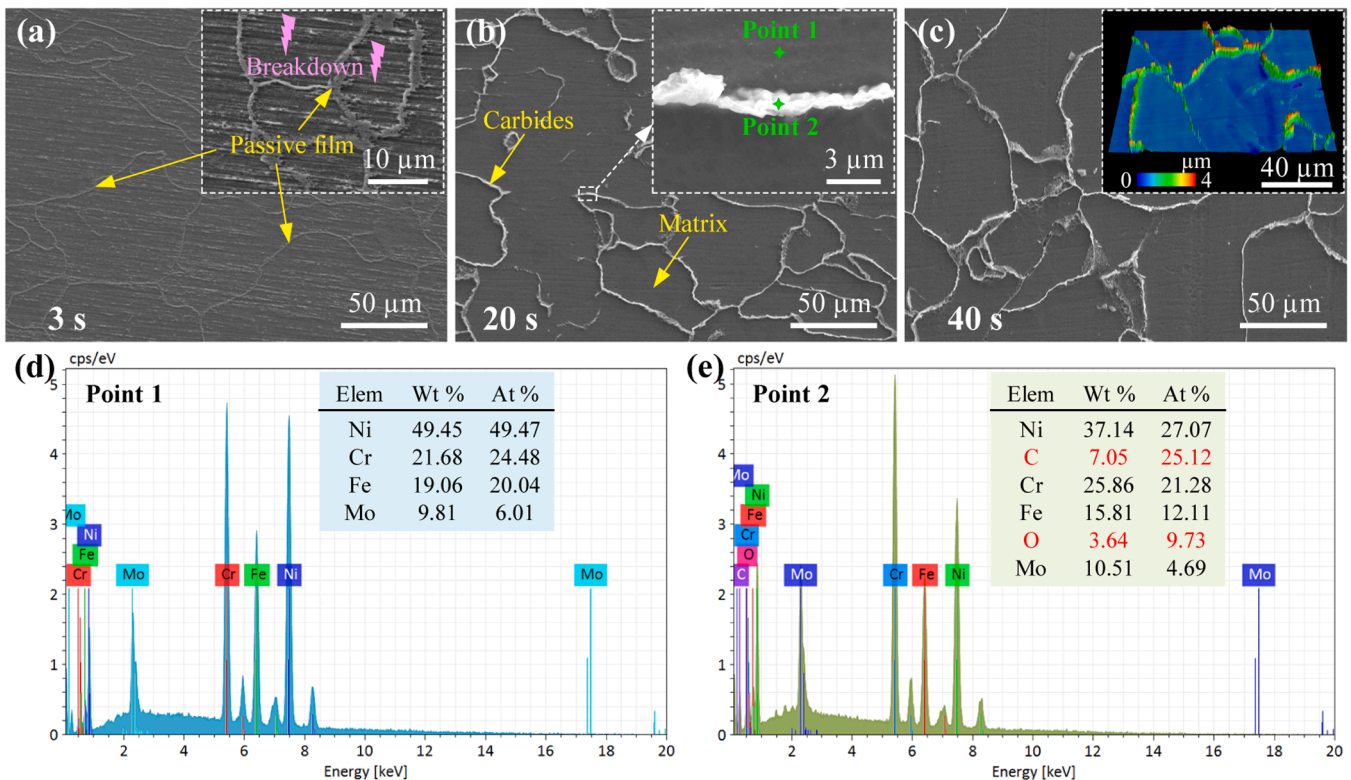


Fig. 6. Dissolved surface morphology of LPBF Hastelloy X in NaCl-aqueous electrolyte for (a) 3 s, (b) 20 s and (c) 40 s; EDS test results for the maker area in (d) Point 1 and (e) Point 2.

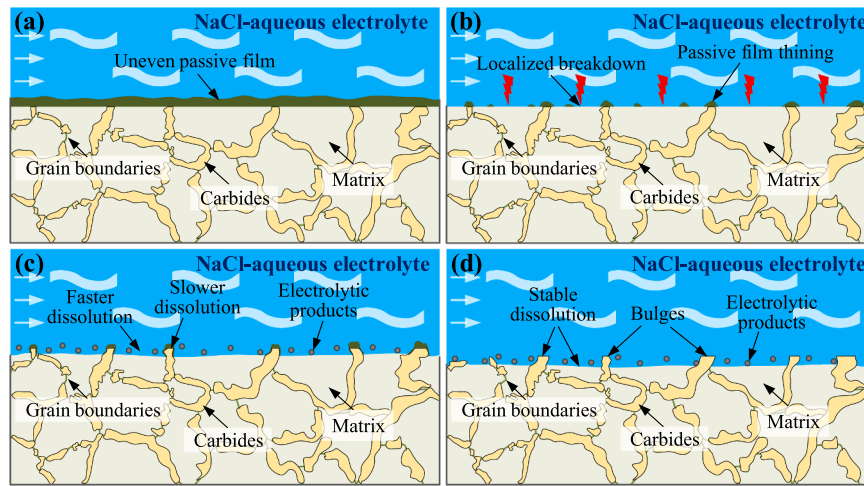


Fig. 7. Schematic of dissolution models of LPBF Hastelloy X in NaCl-aqueous electrolyte: (a) Formation of non-uniform passive film; (b) Local breakdown of passive film; (c) Uneven dissolution; (d) Stable dissolution.

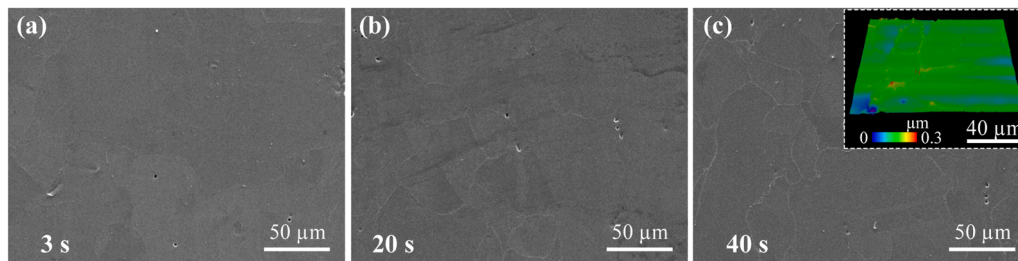


Fig. 8. Dissolved surface morphology of LPBF Hastelloy X in NaCl-EG electrolyte for (a) 3 s, (b) 20 s and (c) 40 s.

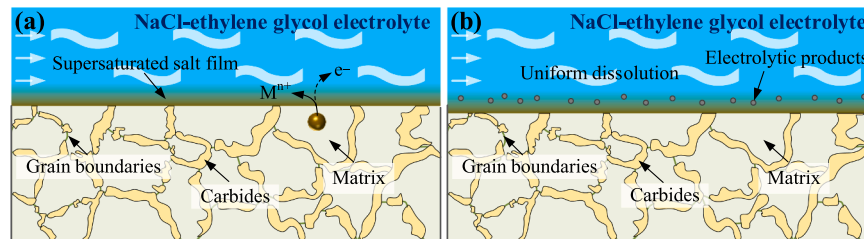
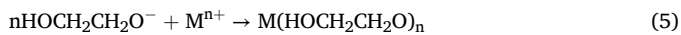
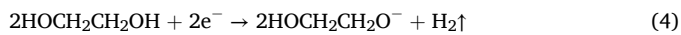


Fig. 9. Schematic of dissolution models of LPBF Hastelloy X in NaCl-EG electrolyte: (a) Formation of supersaturated salt film; (b) Uniform dissolution.

film or electric potential. Consequently, both the matrix and carbides dissolve uniformly in NaCl-EG electrolyte, ultimately yielding a flat and smooth dissolved surface (Fig. 9b).



3.2.3. Effect of current density on dissolution behavior

The current efficiency curves of LPBF Hastelloy X in different electrolytes are presented in Fig. 10. It can be observed that the current efficiency of LPBF Hastelloy X in both electrolytes follows a consistent trend: as the current density increases, the current efficiency initially decreases before stabilizing. Notably, the high current efficiency at low current densities may induce severe stray corrosion during ECP. Within the current density range of 1–4 A·cm⁻², the current efficiency in NaCl-EG electrolyte is significantly lower than that in NaCl-aqueous

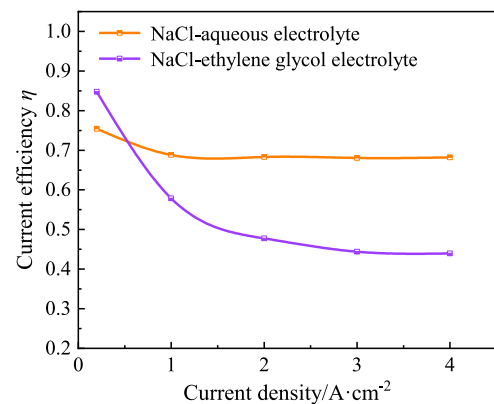


Fig. 10. Current efficiency curves of LPBF Hastelloy X in NaCl-aqueous and NaCl-EG electrolytes.

electrolyte. This phenomenon may be attributed to the oxidative decomposition of ethylene glycol under high current conditions, which consumes partial charges [28]. This reaction becomes more pronounced with increasing current density, consequently reducing the current efficiency. Furthermore, the current efficiency is less than 1 in both NaCl-aqueous and NaCl-EG electrolytes. The primary reason for this observation is that during ECP, some metal ions dissolve in higher valence states, resulting in an actual dissolution quantity that is less than the theoretical value.

Fig. 11 presents the dissolved surface morphologies at different current densities. In NaCl-aqueous electrolyte, the surface quality improves significantly with increasing current density. At a current density of $0.2 \text{ A}\cdot\text{cm}^{-2}$, severe intergranular corrosion is observed, resulting in a surface roughness S_a as high as $5.85 \mu\text{m}$, as shown in Fig. 11a. Yin et al. [26] measured the potential difference between carbides and the matrix, revealing a maximum potential difference of $\sim 78 \text{ mV}$ at carbide-matrix grain boundaries. This potential disparity establishes micro-galvanic effect where the matrix (with more negative potential) undergoes preferential corrosion, causing the carbides to physically detach from the matrix. Consequently, no carbides are observed at $0.2 \text{ A}\cdot\text{cm}^{-2}$. Furthermore, as intergranular corrosion progresses, partially dissolved grains (marked by cyan dashed lines in Fig. 11a) detach from the matrix under electrolyte flow, explaining the highest current efficiency at this current density. When the current density increases to $1 \text{ A}\cdot\text{cm}^{-2}$, the dissolved surface quality improves, as shown in Fig. 11b. This is because at higher applied voltages, the micro-galvanic effect weakens or even disappears, thus causing a change in the electrochemical dissolution behavior. The specific dissolution process has been explained in Section 3.2.2. At $20 \text{ A}\cdot\text{cm}^{-2}$, both carbides and matrix dissolve more uniformly, achieving a smoother surface with roughness S_a reduced to $0.22 \mu\text{m}$, as shown in Fig. 11c.

Fig. 11d demonstrates that the dissolved surface morphology of LPBF Hastelloy X in NaCl-EG electrolyte at $0.2 \text{ A}\cdot\text{cm}^{-2}$ exhibits similar characteristics to that observed in NaCl-aqueous electrolyte. This phenomenon arises because the relatively low metal ion concentration near the anode surface at this current density is insufficient to facilitate the formation of a supersaturated salt film, consequently resulting in non-uniform electrochemical dissolution. When the current density increases to $1 \text{ A}\cdot\text{cm}^{-2}$, the establishment of a supersaturated salt film significantly improves surface quality, reducing the surface roughness S_a to $0.12 \mu\text{m}$, as shown in Fig. 11e. This observation indicates that

NaCl-EG electrolyte enable the attainment of microscopically smoother polished surfaces at lower current densities compared to NaCl-aqueous electrolyte. Further increasing the current density to $4 \text{ A}\cdot\text{cm}^{-2}$ shows negligible changes in both surface morphology and roughness, as shown in Fig. 11f.

3.3. Electrochemical polishing for internal surfaces of channels

3.3.1. Effect of voltage on surface roughness and material removal rate (MRR)

The polishing voltage constitutes one of the most critical process parameters governing both surface quality and material removal efficiency. Fig. 12a illustrates the evolution of surface roughness S_a and MRR for straight channel samples in NaCl-aqueous electrolyte as a function of applied voltage. With increasing voltage, the surface roughness S_a initially decreases before stabilizing, whereas the MRR exhibits a continuous linear increase. 3D topographies of polished surfaces at distinct voltages are presented in Fig. 12b–e. Within the $5\text{--}25 \text{ V}$ range (Fig. 12b–d), the gradual reduction in surface roughness is attributed to enhanced electrochemical dissolution uniformity of LPBF Hastelloy X, as discussed in Section 3.2.3. At voltages exceeding 25 V , S_a stabilizes at approximately $1.4 \mu\text{m}$ without further reduction, as shown in Fig. 12e. This is because ECP is a non-contact polishing method, and the geometric features of the original substrate will be inherited to the polished surface to some extent. Beyond 25 V , surface roughness becomes predominantly governed by geometric profile inheritance, wherein the uniform dissolution of carbides and matrix (microscale homogeneity) cannot further reduce S_a .

In contrast to observations in NaCl-aqueous electrolyte, the surface roughness S_a in NaCl-EG electrolyte remains essentially stable within the $20\text{--}40 \text{ V}$ range, followed by an increase at higher voltages, as depicted in Fig. 13a. The 3D topographies of polished surfaces at varying voltages are presented in Fig. 13b–e. At 20 V and 40 V (Fig. 13b and c), surfaces exhibit smooth morphology with S_a of approximately $1.4 \mu\text{m}$ due to the action of a supersaturated salt film layer. However, elevating the voltage to 60 V (Fig. 13d) induces flow marks aligned with electrolyte flow direction, increasing S_a to $1.67 \mu\text{m}$. This phenomenon arises because Joule heating at elevated voltages reduces ethylene glycol viscosity within the interelectrode gap. When fresh electrolyte enters this region, viscosity gradients trigger perturbation of flow field, thereby generating surface striations. At 80 V , streaks intensify and S_a further

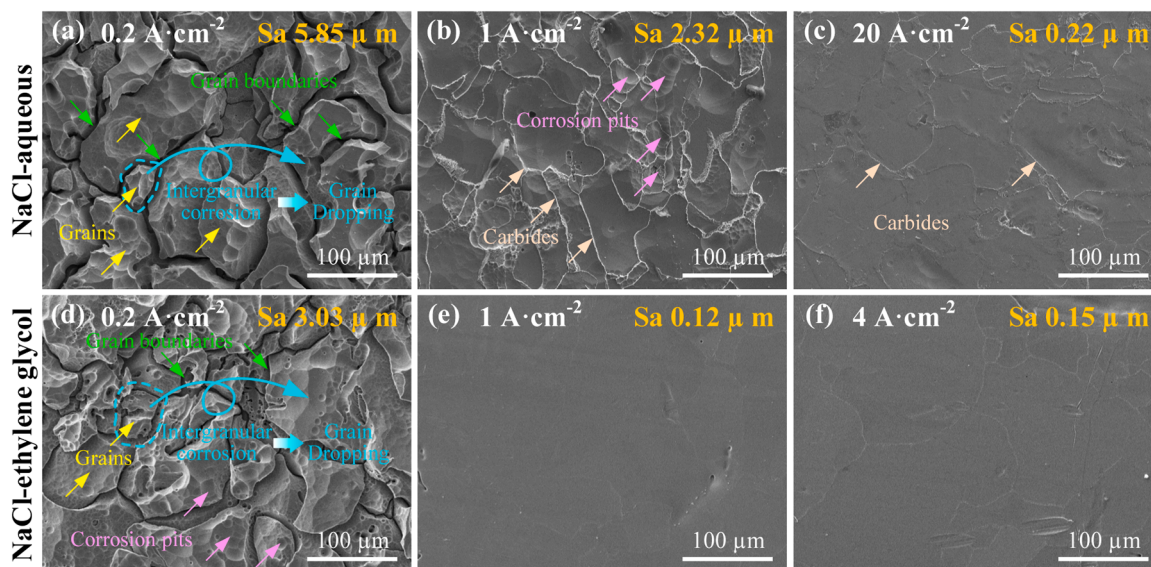


Fig. 11. Dissolved surface morphologies of LPBF Hastelloy X in NaCl-aqueous electrolyte at current densities of (a) $0.2 \text{ A}\cdot\text{cm}^{-2}$, (b) $1 \text{ A}\cdot\text{cm}^{-2}$ and (c) $20 \text{ A}\cdot\text{cm}^{-2}$; Dissolved surface morphologies of LPBF Hastelloy X in NaCl-EG electrolyte at current densities of (d) $0.2 \text{ A}\cdot\text{cm}^{-2}$, (e) $1 \text{ A}\cdot\text{cm}^{-2}$ and (f) $4 \text{ A}\cdot\text{cm}^{-2}$.

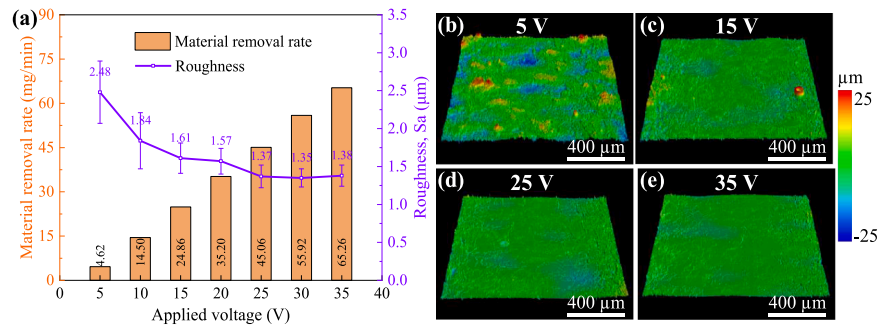


Fig. 12. (a) Evolution of surface roughness Sa and MRR with polishing voltage in NaCl-aqueous electrolyte; 3D topographies of polished surfaces at applied voltage of (b) 5 V, (c) 15 V, (d) 25 V, (e) 35 V.

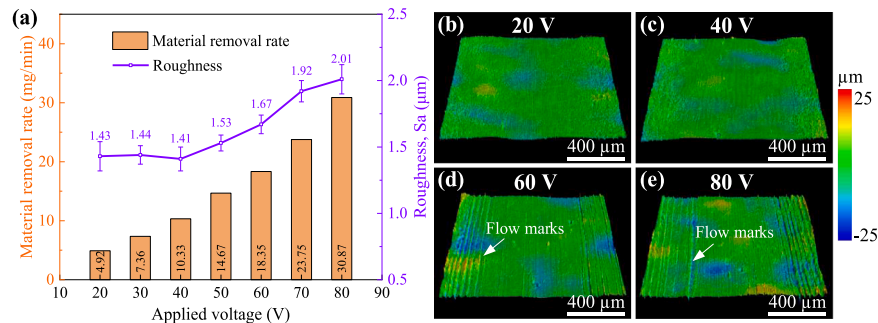


Fig. 13. Evolution of surface roughness Sa and MRR with polishing voltage in NaCl-EG electrolyte; 3D topographies of polished surfaces at applied voltage of (b) 20 V, (c) 40 V, (d) 60 V, (e) 80 V.

rises to 2.01 μm, as shown in Fig. 13e. Consequently, optimal voltage for polishing LPBF Hastelloy X in NaCl-EG electrolyte is 40 V, balancing surface quality and MRR. Nevertheless, compared to NaCl-aqueous electrolyte, NaCl-EG electrolyte yields lower MRR. While 25 V (in NaCl-aqueous electrolyte) and 40 V (in NaCl-EG electrolyte) produce comparable Sa values, the MRR in NaCl-aqueous electrolyte exceeds that in NaCl-EG electrolyte by 336.21 %. Therefore, subsequent polishing

experiments employed NaCl-aqueous electrolyte. Notably, NaCl-EG electrolyte remains preferable when LPBF Hastelloy X components require ultra-high micro-smoothness.

3.3.2. Effect of polishing time on surface roughness and thickness reduction

Fig. 14a displays the surface roughness Sa and corresponding thickness reduction of straight channel samples after ECP at 25 V for

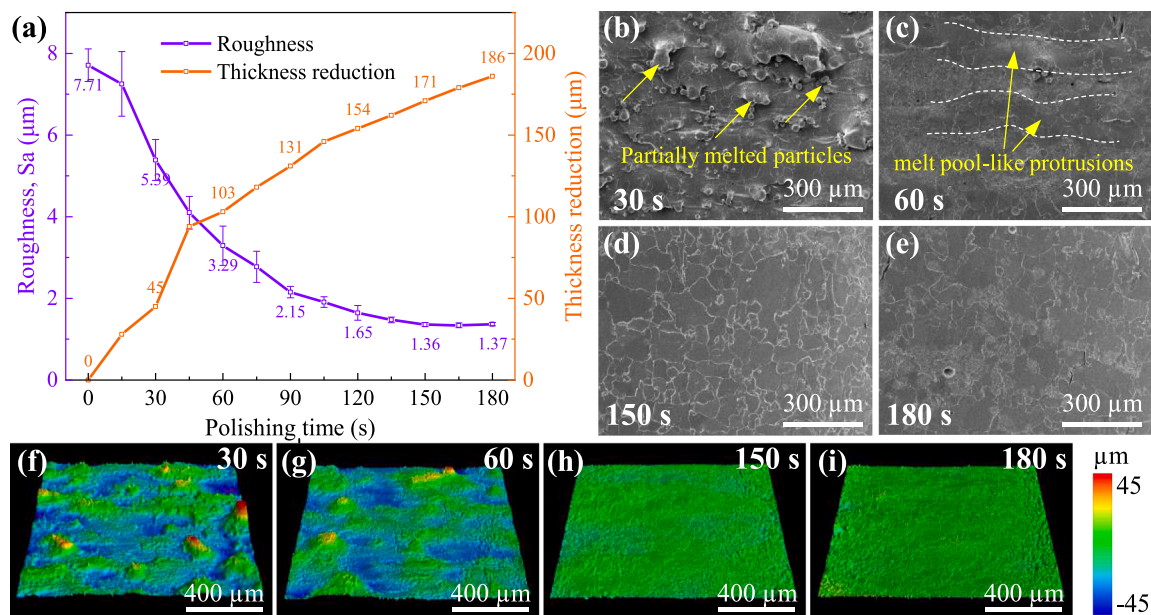


Fig. 14. (a) Variation of surface roughness Sa and thickness reduction with polishing time; SEM images of polished surface at polishing time of (b) 30 s, (c) 60 s, (d) 150 s, (e) 180 s; 3D topographies of polished surface at polishing time of (f) 30 s, (g) 60 s, (h) 150 s, (i) 180 s.

varying durations. Based on the evolution of surface roughness, the ECP process is categorized into three distinct stages: Stage I (0–60 s) exhibits rapid reduction in roughness; Stage II (60–150 s) demonstrates gradual diminishment of roughness; and Stage III (> 150 s) reveals stabilized roughness with negligible variation.

During Stage I, at a polishing duration of 30 s, partially melted particles undergo preferential dissolution due to electric field concentration effects. However, insufficient polishing time prevents complete removal of larger partially melted particles, as illustrated in Fig. 14b and f. Upon extending polishing to 60 s, these particles are fully eliminated, revealing pronounced protruding melt pool structures, while surface roughness S_a decreases to $3.29\text{ }\mu\text{m}$, as shown in Fig. 14c and g. At this juncture, surface roughness is exclusively governed by melt pool-like protrusions. Thus, the rapid roughness reduction in Stage I is attributed to the removal of partially melted particles. During Stage II, prolonged polishing duration facilitates progressive leveling of protruding melt pool structures, thereby promoting further reduction in surface roughness. As depicted in Fig. 14d and h, at a polishing duration of 150 s, melt pool protrusions are substantially leveled, concomitant with a decrease in surface roughness S_a to $1.36\text{ }\mu\text{m}$. In Stage III, prolonged polishing duration fails to further reduce surface roughness S_a , which stabilizes at approximately $1.4\text{ }\mu\text{m}$ due to topographical heredity from the original morphology, as evidenced in Fig. 14e and i. However, thickness reduction continues to increase with extended polishing time. To achieve minimal surface roughness while minimizing thickness reduction for ensuring dimensional accuracy of components, the polishing duration should be controlled at 150 s, where the sample exhibits a thickness reduction of $171\text{ }\mu\text{m}$.

The aforementioned results demonstrate that the achievable limiting value of surface roughness S_a through ECP for improving the surface quality of LPBF Hastelloy X is $\sim 1.4\text{ }\mu\text{m}$. Certainly, this value may vary under specific conditions, as the deposition angle of components influences the initial surface roughness, thereby subsequently affecting the polished surface roughness [31]. Nevertheless, the influence pattern of process parameters on surface roughness remains unchanged.

3.3.3. ECP for curved channels

To evaluate the feasibility of the proposed conformal tool electrode integrated with the workpiece for polishing the surface of LPBF complex components, ECP was performed on the curved channel sample illustrated in Fig. 1c. Given the extensive polishing area, pulsed voltage mode was employed to reduce the average current during polishing and enhance mass transfer efficiency, utilizing the process parameters specified in Table 2.

The initial surface morphology of the curved channel sample is illustrated in Fig. 15a. Due to variations in the curved surface deposition angle, the concentration distribution of partially melted particles becomes non-uniform, resulting in surface roughness S_a ranging from 7.16 to $11.91\text{ }\mu\text{m}$ and poor surface quality uniformity. Fig. 15b demonstrates the surface morphology after ECP, wherein partially melted particles are entirely eliminated and protruding folds are substantially leveled. The S_a decreases significantly to 1.63 – $1.95\text{ }\mu\text{m}$, accompanied by improved surface uniformity. The thickness reduction is approximately $158\text{ }\mu\text{m}$. It should be noted that the combination of process parameters listed in Table 2 is employed solely to validate the effectiveness of the proposed methodology. It is foreseeable that further enhancement in polished

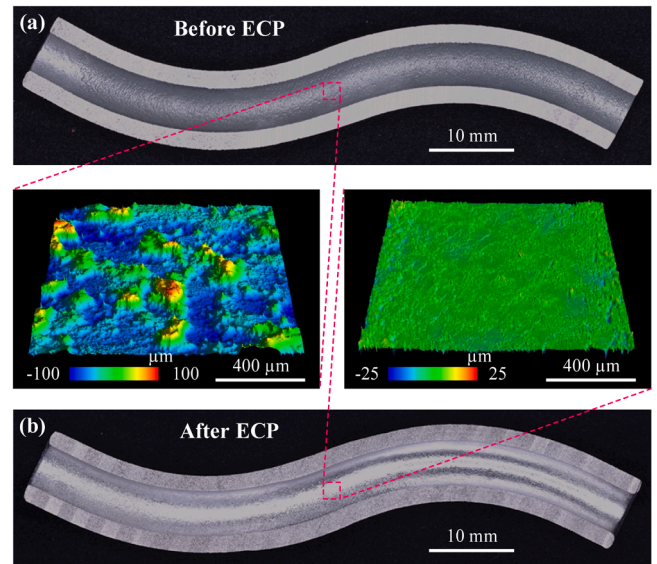


Fig. 15. Surface morphologies of the LPBF Hastelloy X curved channel (a) before and (b) after ECP.

surface quality can be anticipated through optimization of ECP parameters for complex curved surfaces or by introducing additional energy fields during electrochemical polishing.

4. Conclusion

This study systematically investigates the improvement effect of ECP on the surface quality of LPBF Hastelloy X, with a focus on comparing the polishing efficacy between NaCl-aqueous and NaCl-EG electrolytes. To address complex internal surface polishing, a conformal tool electrode is proposed to enhance process accessibility, with the main conclusions summarized as follows:

1. The formation of microscale uneven surface topography occurs in NaCl-aqueous electrolyte due to differential electrochemical dissolution rates between carbides and matrix. This heterogeneous dissolution behavior is governed predominantly by the potential disparity across biphasic interfaces and the stability of passive films, where dissolution uniformity can be enhanced through elevation of current density.
2. In NaCl-EG electrolyte, when the current density exceeds a threshold value, the generation rate of metal ions surpasses their diffusion rate, resulting in the formation of a supersaturated salt film on the anode surface. This film achieves microscopically leveled surfaces by inhibiting dissolution rate disparities between distinct phases.
3. Compared with NaCl-EG electrolyte, the NaCl-aqueous electrolyte enhances MRR by at least 336.21 % while achieving comparable surface roughness. Under 25 V applied voltage and 150 s polishing duration, the sample's surface roughness is reduced by 82 % with a material removal depth of $171\text{ }\mu\text{m}$.
4. To address the challenge of polishing internal cavity surfaces in LPBF components, this study proposes synchronous formation of a conformal tool electrode during part manufacturing. This approach achieves complete removal of partially melted particles from curved channel surfaces, reducing surface roughness S_a from 7.16 to $11.91\text{ }\mu\text{m}$ to 1.63 – $1.95\text{ }\mu\text{m}$. The methodology provides a feasible solution for improving surface quality in additively manufactured internal channels and profiled cavities.

Table 2

Processing parameters of ECP for curved channel.

Parameters	Values
Electrolyte	10 wt% NaCl-aqueous
Voltage	25 V
Pulse frequency	1000 Hz
Duty ratio	10 %
Polishing time	20 min

CRediT authorship contribution statement

zeng yongbin: Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Xiaoyun Hu:** Writing – review & editing, Validation, Supervision. **Yanliang Li:** Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Weiyue Hu:** Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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